content-based and collaborative filtering. Another problem is that of Scalability, when there are simply a huge number of products to choose from, filtering and processing a recommendation list will consume not only time but also computational power. Excess time taken would lead to poor page load time and an increased computational power affects the working of the entire website and other sites which may be hosted on and sharing the same resources. These problems have led to a change in the kind of approaches towards making a recommender system and thus, arose the hybrid system.

Hybrid filtering

Needless to say, this method works towards overcoming the drawbacks of the aforementioned methods. A hybrid system can be more accurate than either of the methods. They can overcome the problem of Cold Start or that of Scalability by using a different algorithm for each scenario.

Algorithms

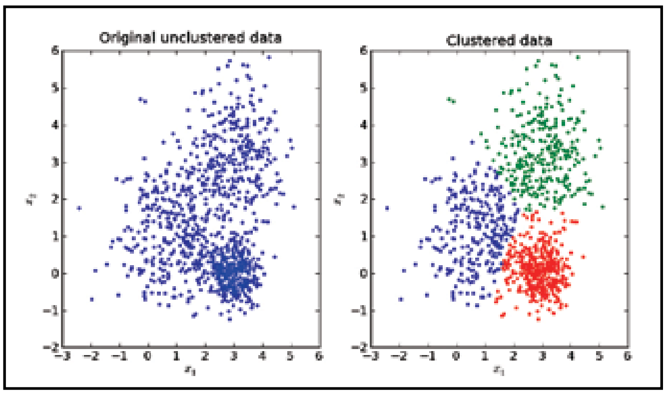
The whole concept comes under machine learning, specifically Collective Intelligence. They make use of multiple algorithms at different stages to generate recommendations. Data Mining algorithms are use to generate the user profile while clustering algorithms are used to shortlist recommendations. Some of the commonly used algorithms are:

- Similarity (Jaccard index & Tanimoto coefficient)
- k Nearest Neighbour algorithm
- n-Dimensional Euclidean distance
- k-means algorithm
- Bayes Theorem
- Singular Value Decomposition

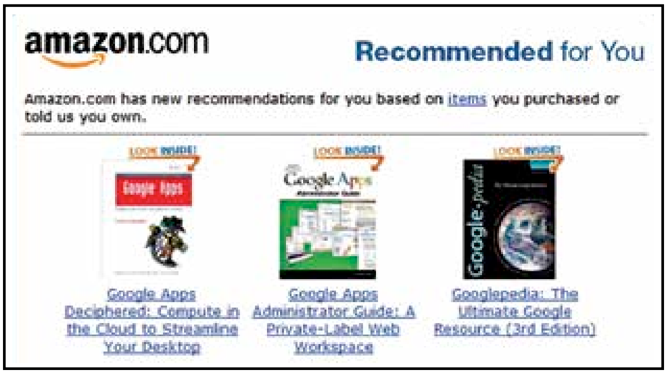
We'll look into one of these algorithms which is very basic and easy to understand. Clustering of data is one of the first steps towards generating categories which can be then used to generate recommendations. A recommendation engine is based on many such algorithms.

K-means clustering algorithm

On the left, we get to see unclustered data. Consider this to be the initial state of all the parameters of every product that the user has looked at so far. This may be a cluster of a hundred products or simply one. Out of these parameters, certain groups may be identified visually but for a computer that is not the case. The algorithm simply associates a few points as the mean. These can



Three clusters obtained using the k-means algorithm



Purchase history affects recommendations more than anything

found at the centre of each coloured cluster as seen on the right. From this mean point, the distance to all data points are calculated and then based on which points are closest, a small cluster is formed. It is quite possible that during the formation of such a cluster, the mean might not be a feasible point and thus it may be moved, removed or even a new mean point could be added. After multiple iterations we obtain the cluster on the right. This data set could be indicative of which products are the most popular and thus are more likely to get recommended to all customers. This way, a cold start problem can be easily resolved.

A combination of many algorithms

A recommender system is by no means simple, it has a lot of different algorithms working towards producing just one result. A commonly known system that used a lot many algorithms was the one used by Netflix back in 2007, called Cinematch. Over 170 algorithms were used for each individual result. However, this was not enough and Netflix introduced a prize for $1,000,000 to anyone who could improve the accuracy of their system by 10%. This may seem to be a minor improvement but that 10% was more than enough to make good for the $1,000,000 prize money they were giving away. Going on a meta level increases the accuracy of a system. So if you were to start describing each parameter with another parameter then another dimension comes into the picture. "User Generated Tags" a feature implemented by Steam, the online game distribution platform is one such thing. It is a form of collaborative filtering where user inputs are random. These tags may be used by other customers and they'll be recommended all the games that have the tags that they used.

How valuable are they?

The answer to this question depends on a lot many parameters. Primarily, it depends on how big your inventory is. Then it comes to breaking down the products into varying categories and then describing them with as many parameters as possible. All of this while maintaining consistency between the parameters. For example, for two different phones describing the screen of one screen using "surface area" and the other by using "diagonal length" gets you no way of associating the two. Then you need to have a big enough inventory to start off. It is a fact that each individual customer is only interested in a small fraction of products that a store offers. Having a recommender system otherwise ends up giving less returns in the long run. However, if none of these issues plagues your enterprise then you might see a surge of up to 60% in sales. Netflix revealed that about